Jounghyun Yoo

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EDUCATION

Pohang University of Science and Technology (POSTECH) Mar. 2014 - Aug. 2019
Ph.D. in Chemical Engineering
Advisor: Prof. Shi-Woo Rhee

Pohang University of Science and Technology (POSTECH) Mar. 2010 - Feb. 2014
B.S. in Chemical Engineering, Magna cum laude

PROFESSIONAL EXPERIENCE

California Institute of Technology (Caltech) Sep. 2021 - Present
Postdoctoral Researcher, Department of Medical Engineering
Advisor: Prof. Wei Gao

Ulsan National Institute of Science and Technology (UNIST) Sep. 2021 - Sep. 2022; Jan. 2024 – Aug. 2026
Postdoctoral Researcher, Department of Biomedical Engineering
Advisor: Prof. Jinmyoung Joo

Korea Institute of Science and Technology (KIST) Sep. 2019 - Aug. 2021
Postdoctoral Researcher, Center for Theragnosis, Biomedical Research Institute
Advisor: Dr. Sehoon Kim

RESEARCH INTERESTS

Medical micro/nanorobotics. Designing micro/nanorobotic systems for active biological transport, barrier penetration, targeted localization, controlled therapeutic delivery, and adaptive interaction with biological microenvironments.

Imaging-guided biofabrication. Developing non-invasive biofabrication strategies to construct functional biomaterial interfaces within living tissues for tissue-integrated monitoring, localized intervention, and regenerative medicine.

Bioactive nanomaterials. Engineering biocompatible and biodegradable nanomaterial platforms with tunable physicochemical properties and biological functions to regulate cellular interactions, biodistribution, imaging contrast, and therapeutic performance.

SELECTED PUBLICATIONS

*†Equal contribution; *Corresponding author.*

1. **J. Yoo**†, S. Kim†, W. Yoo, H. Han, G. Kim, A. W. Tessema, S. Kim, S. Tang, J. Kim, H. D. Truong, X. Ma, G. Kim, S. Won, J.-W. Choi, J. Hwang, J. Oh, Y. Seok, J. K. Kim, H. Cho, T.-E. Park, J. Joo, W. Gao, Photothermally gated nanorobots for active blood-brain barrier transcytosis and brain delivery. *Science Robotics*, accepted.
2. H. Han†, X. Ma†, **J. Yoo**†, S. Tang, H. D. Truong, P. Serles, K. Zhao, E. Criado-Hidalgo, D. D. Tang, Y.-J. Kim, G. Chen, J. Qu, J. Li, C. Fezzi, J. Zhang, C. Hays, R. J. Reyes Segura, Z. Jin, A. Liu, M. G. Shapiro, J. R. Greer, Q. Zhou, W. Gao, Ultrasound-propelled microrobots for therapeutic gas delivery and gas–sonodynamic therapy. *Nature Communications*, accepted.
3. **J. Yoo**, J. Joo, W. Gao, Bridging the barrier: Micro- and nanorobotics for active brain delivery. *Matter*, 9, 102858 (2026).
4. M. Kim, K. Kim, J. Lee, S. Lee, S. Choi, S. A. Park, E. Jeong, S.-Y. Choi, H. H. Park, T.-E. Park, T. Kwon, K. Myung, **J. Yoo***, S. W. Cho*, J. Joo*, FAST-CRISPR: Fusogenic association and secured transfection of CRISPR/Cas9 ribonucleoproteins using lipid-silica hybrid nanoparticles for therapeutic genome editing. *Small*, 22, e11362 (2026).
5. S. Lee†, **J. Yoo**†, G. Bae, R. Thangam, J. Heo, J. Y. Park, H. Choi, C. Kim, J. An, J. Kim, K. R. Mun, S. Shin, K. Zhang, P. Zhao, Y. Kim, N. Kang, S.-B. Han, D. Kim, J. Yoon, M. Kang, J. Kim, L. Yang, S. Karamikamkar, J. Kim, Y. Zhu, A. H. Najafabadi, G. Song, D.-H. Kim, K.-B. Lee, S. J. Oh, H.-D. Jung, H.-C. Song, W. Y. Jang, L. Bian, Z. Chu, J. Yoon, J. S. Kim, Y. S. Zhang, Y. Kim, H. S. Jang, S. Kim, H. Kang, Photonic control of ligand nanospacing in self-assembly regulates stem cell fate. *Bioactive Materials*, 34, 164 (2024).
6. **J. Yoo**†, S. Tang†, W. Gao, Micro- and nanorobots for biomedical applications in the brain. *Nature Reviews Bioengineering*, 1, 308 (2023).

PUBLICATIONS

*†Equal contribution; *Corresponding author.*

1. **J. Yoo**†, S. Kim†, W. Yoo, H. Han, G. Kim, A. W. Tessema, S. Kim, S. Tang, J. Kim, H. D. Truong, X. Ma, G. Kim, S. Won, J.-W. Choi, J. Hwang, J. Oh, Y. Seok, J. K. Kim, H. Cho, T.-E. Park, J. Joo, W. Gao, Photothermally gated nanorobots for active blood-brain barrier transcytosis and brain delivery. *Science Robotics*, accepted.
2. H. Han†, X. Ma†, **J. Yoo**†, S. Tang, H. D. Truong, P. Serles, K. Zhao, E. Criado-Hidalgo, D. D. Tang, Y.-J. Kim, G. Chen, J. Qu, J. Li, C. Fezzi, J. Zhang, C. Hays, R. J. Reyes Segura, Z. Jin, A. Liu, M. G. Shapiro, J. R. Greer, Q. Zhou, W. Gao, Ultrasound-propelled microrobots for therapeutic gas delivery and gas–sonodynamic therapy. *Nature Communications*, accepted.
3. **J. Yoo**, J. Joo, W. Gao, Bridging the barrier: Micro- and nanorobotics for active brain delivery. *Matter*, 9, 102858 (2026).
4. S. Tang, H. Han, X. Ma, P. N. Patel, C. Gong, J. Zhang, E. Criado-Hidalgo, **J. Yoo**, J. Li, G. Kim, S. Yin, D. Wu, M. G. Shapiro, Q. Zhou, W. Gao, Enzymatic microbubble robots. *Nature Nanotechnology*, 21, 397 (2026).
5. M. Kim, K. Kim, J. Lee, S. Lee, S. Choi, S. A. Park, E. Jeong, S.-Y. Choi, H. H. Park, T.-E. Park, T. Kwon, K. Myung, **J. Yoo***, S. W. Cho*, J. Joo*, FAST-CRISPR: Fusogenic association and secured transfection of CRISPR/Cas9 ribonucleoproteins using lipid-silica hybrid nanoparticles for therapeutic genome editing. *Small*, 22, e11362 (2026). Featured on Journal Cover.
6. **J. Yoo**†, Y. Kim†, J. H. Back†, J. Shin, P. K. Bae, K. M. Park, M. Kim, Y. H. Seo, Y. Bak, Y. H. Heo, J. Heo, H. Choi, Y. Kim, S. Lee, J. E. Lee, S. Jeong, J.-K. Yang, S. Kim, Surface-engineered nanobeads for regioselective antibody binding: A robust immunoassay platform leveraging catalytic signal amplification. *Biosensors and Bioelectronics*, 281, 117463 (2025).
7. E. Davoodi, J. Li, X. Ma, A. H. Najafabadi, **J. Yoo**, G. Lu, E. S. Sani, S. Lee, H. Montazerian, G. Kim, J. Williams, J. W. Yang, Y. Zeng, L. S. Li, Z. Jin, B. Sadri, S. S. Nia, L. V. Wang, T. K. Hsiai, P. S. Weiss, Q. Zhou, A. Khademhosseini, D. Wu, M. G. Shapiro, W. Gao, Imaging-guided deep tissue in vivo sound printing. *Science*, 388, 616 (2025).
8. Y. Choi, P. Jin, S. Lee, Y. Song, R. Y. Tay, G. Kim, **J. Yoo**, H. Han, J. Yeom, J. H. Cho, D. Kim, W. Gao, All-printed chip-less wearable neuromorphic system for multimodal physicochemical health monitoring. *Nature Communications*, 16, 5689 (2025).
9. J. Seong, M. Kim, **J. Yoo**, D. L. Mack, J. H. Lee, J. Joo, Sustained release of HIF-2α inhibitors using biodegradable porous silicon carriers for enhanced immunogenic cell death of malignant Merkel cell carcinoma. *ACS Applied Materials & Interfaces*, 17, 7449 (2025).
10. S. Lee†, **J. Yoo**†, G. Bae, R. Thangam, J. Heo, J. Y. Park, H. Choi, C. Kim, J. An, J. Kim, K. R. Mun, S. Shin, K. Zhang, P. Zhao, Y. Kim, N. Kang, S.-B. Han, D. Kim, J. Yoon, M. Kang, J. Kim, L. Yang, S. Karamikamkar, J. Kim, Y. Zhu, A. H. Najafabadi, G. Song, D.-H. Kim, K.-B. Lee, S. J. Oh, H.-D. Jung, H.-C. Song, W. Y. Jang, L. Bian, Z. Chu, J. Yoon, J. S. Kim, Y. S. Zhang, Y. Kim, H. S. Jang, S. Kim, H. Kang, Photonic control of ligand nanospacing in self-assembly regulates stem cell fate. *Bioactive Materials*, 34, 164 (2024).
11. H. Han, X. Ma, W. Deng, J. Zhang, S. Tang, O. S. Pak, L. Zhu, E. Criado-Hidalgo, C. Gong, E. Karshalev, **J. Yoo**, M. You, A. Liu, C. Wang, H. K. Shen, P. N. Patel, C. L. Hays, P. J. Gunnarson, L. Li, Y. Zhang, J. O. Dabiri, L. V. Wang, M. G. Shapiro, D. Wu, Q. Zhou, J. R. Greer, W. Gao, Imaging-guided bioresorbable acoustic hydrogel microrobots. *Science Robotics*, 9, eadp3593 (2024).
12. S. Tang, D. Tang, H. Zhou, Y. Li, D. Zhou, X. Peng, C. Ren, Y. Su, S. Zhang, H. Zheng, F. Wan, **J. Yoo**, H. Han, X. Ma, W. Gao, S. Wu, Bacterial outer membrane vesicle nanorobot. *Proceedings of the National Academy of Sciences of the United States of America*, 121, e2403460121 (2024).
13. H. Chang, Y. Park, K. Kim, C. Han, Y. Yoon, W. Yoo, **J. Yoo**, D. Lee, H. Han, K. Kim, J. Joo, W. Kwon, Room-temperature phosphorescence of defect-engineered silica nanoparticles for high-contrast afterglow bioimaging. *Chemical Engineering Journal*, 493, 152529 (2024).
14. **J. Yoo**†, S. Tang†, W. Gao, Micro- and nanorobots for biomedical applications in the brain. *Nature Reviews Bioengineering*, 1, 308 (2023).
15. **J. Yoo**†, K. Kim†, S. Kim, H. H. Park, H. Shin, J. Joo, Tailored polyethylene glycol grafting on porous nanoparticles for enhanced targeting and intracellular siRNA delivery. *Nanoscale*, 14, 14482 (2022). Featured on Journal Cover.
16. T. F. Anjong†, H. Choi†, **J. Yoo**†, Y. Bak, Y. Cho, D. Kim, S. Lee, K. Lee, B.-G. Kim, S. Kim, Multifunction-harnessed afterglow nanosensor for molecular imaging of acute kidney injury in vivo. *Small*, 18, 2200245 (2022).
17. Y. Kim, R. Thangam, **J. Yoo**, J. Heo, J. Y. Park, N. Kang, S. Lee, J. Yoon, K. R. Mun, M. Kang, S. Min, S. Y. Kim, S. Son, J. Kim, H. Hong, G. Bae, K. Kim, S. Lee, L. Yang, J. Y. Lee, J. Kim, S. Park, D.-H. Kim, K.-B. Lee, W. Y. Jang, B. H. Kim, R. Paulmurugan, S.-W. Cho, H.-C. Song, S. J. Kang, W. Sun, Y. Zhu, J. Lee, H.-J. Kim, H. S. Jang, J. S. Kim, A.

Khademhosseini, Y. Kim, S. Kim, H. Kang, Photoswitchable microgels for dynamic macrophage modulation. *Advanced Materials*, 34, e2205498 (2022). Featured on Journal Cover.

18. D. Jang, S. Yu, K. Chung, **J. Yoo**, F. M. Mota, J. Wang, D. J. Ahn, S. Kim, D. H. Kim, Direct deposition of anatase TiO_2 on thermally unstable gold nanobipyramid: Morphology-conserved plasmonic nanohybrid for combinational photothermal and photocatalytic cancer therapy. *Applied Materials Today*, 27, 101472 (2022).
19. Y.-D. Lee†, H. J. Shin†, **J. Yoo**†, G. Kim, M.-K. Kang, J. J. Lee, J. Bang, J.-K. Yang, S. Kim, Metal complexation-mediated stable and biocompatible nanoformulation of clinically approved near-infrared absorber for improved tumor targeting and photonic theranostics. *Nano Convergence*, 8, 36 (2021).
20. K. Jeong, D. Kim, H. J. Kim, Y.-D. Lee, **J. Yoo**, D. Jang, S. Lee, H. Park, Y. Kim, A. Singh, D. J. Ahn, D. H. Kim, J. Bang, J. Kim, P. N. Prasad, S. Kim, Photoechogenic inflatable nanohybrids for upconversion-mediated sonotheranostics. *ACS Nano*, 15, 18394 (2021).
21. J. Shin, D. M. Kang, **J. Yoo**, J. Heo, K. Jeong, J. H. Chung, Y. S. Han, S. Kim, Superoxide-responsive fluorogenic molecular probes for optical bioimaging of neurodegenerative events in Alzheimer's disease. *Analyst*, 146, 4748 (2021). Featured on Journal Cover.
22. **J. Yoo**†, H. Kim†, H. Chang†, W. Park, S. K. Hahn, W. Kwon, Biocompatible organosilica nanoparticles with self-encapsulated phenyl motifs for effective UV protection. *ACS Applied Materials & Interfaces*, 12, 9062 (2020).
23. **J. Yoo**†, U. Ryu†, W. Kwon, K. M. Choi, A multi-dye containing MOF for the ratiometric detection and simultaneous removal of $Cr_2O_7^{2-}$ in the presence of interfering ions. *Sensors and Actuators B: Chemical*, 283, 426 (2019).
24. Y. Park†, **J. Yoo**†, M.-H. Kang†, W. Kwon, J. Joo, Photoluminescent and biodegradable porous silicon nanoparticles for biomedical imaging. *Journal of Materials Chemistry B*, 7, 6271 (2019). Featured on Journal Cover.
25. H. Chang†, **J. Yoo**†, H. Kim, W. Park, S. K. Hahn, W. Kwon, Controlled growth of fluorescent silica nanoparticles using two-phase orthogonal solvents for bioimaging. *Journal of Luminescence*, 214, 116529 (2019).
26. T. Lee, H. T. Bui, **J. Yoo**, M. Ra, S. H. Han, W. Kim, W. Kwon, Formation of TiO_2@carbon core/shell nanocomposites from a single molecular layer of aromatic compounds for photocatalytic hydrogen peroxide generation. *ACS Applied Materials & Interfaces*, 11, 41196 (2019).
27. **J. Yoo**†, S. Han†, W. Park, T. Lee, Y. Park, H. Chang, S. K. Hahn, W. Kwon, Defect-induced fluorescence of silica nanoparticles for bioimaging applications. *ACS Applied Materials & Interfaces*, 10, 44247 (2018).
28. D. Lee†, S. Beack†, **J. Yoo**†, S.-K. Kim, C. Lee, W. Kwon, S. K. Hahn, C. Kim, In vivo photoacoustic imaging of livers using biodegradable hyaluronic acid-conjugated silica nanoparticles. *Advanced Functional Materials*, 28, 1800941 (2018). Featured on Journal Cover.
29. U. Ryu†, **J. Yoo**†, W. Kwon, K. M. Choi, Tailoring nanocrystalline metal-organic frameworks as fluorescent dye carriers for bioimaging. *Inorganic Chemistry*, 56, 12859 (2017).
30. Y. Park†, **J. Yoo**†, B. Lim, W. Kwon, S.-W. Rhee, Improving the functionality of carbon nanodots: Doping and surface functionalization. *Journal of Materials Chemistry A*, 4, 11582 (2016).

MANUSCRIPTS IN PREPARATION

1. **J. Yoo**, D.-M. Lee, J. Bang, Y. E. Cho, Q. He, Q. Zhou, W. Gao, Imaging-guided in vivo biofabrication for functional interfacing with dynamic tissues. In preparation.

PATENTS

1. H. Kang, S. Kim, H. S. Jang, Y. Kim, **J. Yoo**, J. Heo, G. Bae, N. Kang, S. Lee, Y. Kim, Self-assembled complex controlled by light energy for stem cells and nanosystem including same, Korean patent (application: 10-2023-0029829, 7 Mar. 2023; granted: 10-2897463-0000, 3 Dec. 2025).
2. W. Kwon, **J. Yoo**, S. K. Hahn, H. Kim, Ultraviolet protection silica nanoparticle composition and its manufacturing method, Korean patent (application: 10-2019-0173419, 23 Dec. 2019; granted: 10-2340304-0000, 13 Dec. 2021).
3. K. M. Choi, W. Kwon, U. Ryu, **J. Yoo**, Imaging-agents loaded nanocrystalline metal-organic frameworks for containing and method for preparing the same, Korean patent (application: 10-2017-0052355, 24 Apr. 2017; granted: 10-1890588-0000, 16 Aug. 2018).

HONORS AND FELLOWSHIPS

UNIST Global Postdoctoral Short-term Training Fellowship Sep. 2025 - Dec. 2025
Overseas postdoctoral research fellowship supported by Ulsan National Institute of Science and Technology (UNIST)

Sejong Science Fellowship (Overseas Training Track) Sep. 2024 - Aug. 2025

Overseas postdoctoral research fellowship supported by the Ministry of Science and ICT (MSIT), Republic of Korea

Korea Health Industry Development Institute (KHIDI) Fellowship Sep. 2021 - Sep. 2022
Overseas postdoctoral research fellowship supported by the Ministry of Health and Welfare, Republic of Korea

Best Paper of the Year Nov. 2018
Awarded by the Department of Chemical Engineering at POSTECH

National Scholarship for Science and Engineering Mar. 2010 - Feb. 2014
Supported by Korea Student Aid Foundation (KOSAF)

TEACHING, MENTORING, AND LEADERSHIP

Postdoctoral Researcher Coordinator, Caltech Jan. 2024 - Sep. 2024
Appointed by the Caltech Korean Graduate Student Association (CKGSA)

Project Leader and Research Mentor, KIST Fall 2019 - Spring 2021
Mentored undergraduate and doctoral researchers in bioactive nanomaterials for biomedical applications

Teaching Assistant, POSTECH Fall 2014 - Spring 2018
Solid State Chemical Engineering (CHEB360)
Physical Chemical Laboratory (CHEB212)
Introduction to Chemical Engineering (CHEB100)

Peer Tutor, POSTECH Spring 2013
Chemical Engineering Thermodynamics (CHEB204)

Student Advisor, POSTECH Spring 2012
Future Planning for College Life I (MSUS102)

REFERENCES

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Prof. Shi-Woo Rhee
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